How Can the [MASK] Know? The Sources and Limitations of Knowledge in BERT

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***Abstract*—We explore the idea of using the pre-trained BERT as a source of factual knowledge, analyze which components of the model are responsible for its ability to answer questions requiring factual knowledge, and study the transferability of the knowledge to downstream tasks. Our experiments show that the Language Modeling Head is indispensable for predicting facts, implying that transferability of any knowledge captured in the model is limited. While the dominant approach to researching how knowledge is stored in language models focuses on tailoring question formulation to optimize the retrieval quality, we find question patterns easily understood by humans that confuse BERT to the point that the answer does not make sense. The nature of the found patterns implies that the stored knowledge is fragile and based on token co-occurrence in the training set used during pre-training, rather than generalization or inference. Moreover, using a novel, hand-crafted dataset we show that BERT is vulnerable to common misconceptions, which could have fatal effects on downstream applications. Overall, we conclude that BERT offers low and unreliable performance out of the box. Jupyter notebooks with experiments are available on GitHub.**[1]

I. Introduction

In the recent years, there has been a surge in natural language processing results that can be attributed to Transformer [1] based models [2], [3], [4]. Studies have shown that the models are able to capture superficial language patterns [5], [6] and queries in natural language can be constructed to retrieve facts about the world from the pre-trained models [7], [8], [9]. While capturing general knowledge in a language model is certainly appealing, it is necessary to understand the source of knowledge and possible confusion to maximize the benefits for downstream applications. Furthermore, while the performance on knowledge benchmarks has progressively improved, there is a lack of understanding of how the Transformer-based models are able to predict facts, what roles different components play, and how robust and transferable the knowledge captured by the model during pre-training is. Addressing these questions is necessary if one likes to use the knowledge in the pre-trained models for critical applications such as the medical domain.

In this paper we attempt to fill these gaps in understanding BERT [3], arguably the most popular and the most successful variation of the Transformer-based models, as a case study. While some of the findings may be specific to BERT, the methodology and most findings are applicable to other Transformer-based models. First, in Section IV-A, we show that BERT's ability to answer queries requiring factual knowledge relies mainly on superficial co-occurrences, rather than generalization and its ability to reason or infer, and can be easily influenced by inconsequential syntax changes (such as removing a dot or [SEP]). For instance, given the input sequence "[CLS] Vaccines cause [MASK]. [SEP]", the top five predictions are: 'cancer', 'disease', 'death', 'diseases', 'mortality'; however, if we change the sentence to "[CLS] Vaccines cause [MASK] [SEP]", the top five predictions become ';', '.', '?', '!', and '|'. Using BERT as a reliable source of knowledge could have catastrophic effects on downstream applications.

Second, in Section IV-C, we show that the *Layer Normalization* (LayerNorm) [10] module in the language model (LM) head, a classifier layer discarded after pre-training, is responsible for BERT's ability to predict facts. Without the LM head, retrieved facts are dominated by [CLS], a special token used in BERT, due to its large magnitude; after swapping the magnitude of 'the' with that of [CLS], 'the' dominates in most cases. It is evident that when LM Head is discarded and only the encoder stack is reused, which is the case for most downstream tasks, the knowledge in BERT cannot be transferred. Note that this does not imply the encoder stack is not useful. As the encoder stack transforms the input into contextualized representations that improve the performance of downstream applications, one can replace the LM head by training a customized one to achieve good performance, such as KG-BERT [11]. As the custom layers are only fine-tuned on specific and typically much smaller dataset(s), they would not have the benefits of much larger dataset(s) used during pre-training.

Third, in Section V, we demonstrate that one word can dramatically alter the predicted answers. We observe that only a few tokens other than the special [MASK] token can be used as a placeholder for the prediction, despite the masking strategy used during pre-training suggesting that the model should be able to correct any token based on its context. We find that most of the words are reconstructed exactly, regardless of the context. This implies the mismatches between pre-training (where [MASK] is used) and applications (no [MASK] token is used typically)[12] could alter the knowledge in BERT substantially. Furthermore, it also implies that BERT knowledge may not generalize well to new input sequences. Overall, the knowledge in BERT lacks the desired robustness and is fragile.

Finally, in Section VI, we have manually constructed a novel misconceptions dataset for evaluation of the robustness and

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[1]Project URL: https://github.com/tenpercent/knowledge-and-confusion

reliability of Language Models trained in a self-supervised manner as knowledge sources; we have shown that BERT does not handle the task well, and we speculate that it could lead to biases and other negative effects on downstream applications.

The rest of this paper is organized as follows. Section II contains the related work. In Section III, we provide background about BERT structure and training objectives. Section IV is devoted to exploration of LM head's role in BERT's token reconstruction capability. Section V comprises demonstration of knowledge fragility by using alternative masking tokens. In Section VI, we study the factually wrong knowledge stored in BERT. We discuss our results in Section VII and conclude with the summary of our results and possible future work directions in Section VIII.

II. Related Work

Dissecting the BERT internals has sparked wide research interests, to the point of forking a deluge of results into a separate subfield coined *BERTology*. Rogers et al. [13] provide a comprehensive survey of that research direction. Petroni et al. [7] design a probe dataset coupled with a framework for common sense and factual knowledge extraction from pre-trained Transformer models [14], [3], [15], [2], [4]. Talmor et al. [16] devise a set of probing tasks involving symbolic reasoning, such that they may potentially be difficult to be captured by the masked language modeling objective, as the latter is designed with the emphasis on capturing tokens' co-occurrences. Yao et al. [11] fine-tune BERT to perform relation prediction and link prediction tasks on knowledge graphs via (entity, relation, entity) triple classification. Jiang et al. [8] explore strategies for constructing prompts in natural language to extract knowledge from language models, as arbitrary requests prove to be too difficult to answer correctly. Petroni et al. [9] study the question context influence on the precision of the result, showing potential for improvement when the additional context relevant to the question is provided, while arguing that knowledge extraction is robust to adding a noisy or irrelevant context. However, they omit discussing why they observe such effect. We believe that the reason is related to co-occurrences of the original questions and additional context tokens. Bosselut et al. [17] introduce a framework for knowledge base (KB) filling via fine-tuning a different transformer, GPT [2]. Note that none of the studies try to understand the underlying mechanisms for the knowledge and confusion in pre-trained BERT, which is the focus of this paper.

Motivated by the lack of guarantees on BERT's reliability on knowledge intensive tasks, efforts to integrate external knowledge bases into neural Language Models have been made. Peters et al. [18] embed and integrate human-crafted knowledge from multiple KBs (i.e., WordNet, Wikipedia) into BERT using self-supervision on unlabeled data, and report improved factual knowledge recall compared to the original BERT model. Guu et al. [19] propose REALM, a language model augmented with retrieval capabilities, allowing the model to retrieve and attend over documents from a knowledge corpus (i.e., Wikipedia). Through utilization of external knowledge

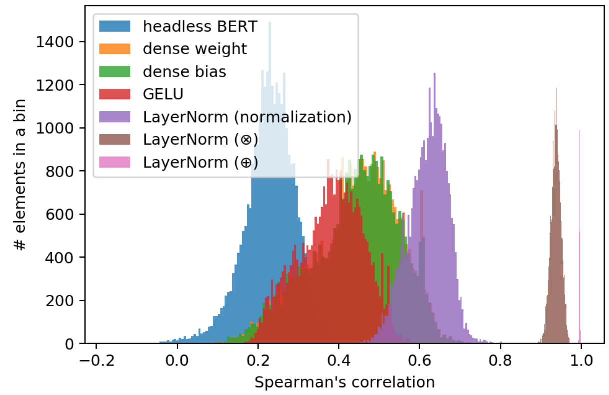

Fig. 1: Histogram of Spearman's rank order correlations between output of BERT with LM head and vectors produced from headless BERT output and each of the LM head activations using inner product with token embeddings, aggregated over the ConceptNet dataset with 29774 samples. We note a sharp increase in similarity after LayerNorm; it coincides with the point when the predictions obtained from a layer within LM head start resembling the predictions of the entire BERT with LM head.

source, [19] are able to improve performance on a number of Open-Domain Question Answering benchmarks. Mulang et al. [20] study the importance of a knowledge base in Named Entity Disambiguation task. Authors show that an RNN model [21], provided with context derived from a KB, outperforms large Transformer-based language models on this task. Although, Transformer-based LMs achieve better performance than the RNN when fed with the context from KB, these results confirm that models' parameters, even when trained on a large corpora, are not a reliable replacement of an explicit knowledge source.

III. Overview of BERT

BERT is a stack of Transformer encoder blocks [1], each composed of two sub-layers: multi-head attention and a position-wise fully connected feed-forward network. In the multi-head attention sub-layer, every input token is linearly projected h times with different, learned parameter matrices for *Query (Q)*, *Key (K)* and *Value (V)*, into d_h dimensional vectors, where h is the number of attention heads and d_h is the attention head dimensionality. The softmax of a scaled inner product between Q and K is multiplied with *(V)* creating weighted, context dependent representations. The representations produced by each head are concatenated and transformed by a linear projection. In the second sub-layer, position-wise, fully connected feed-forward network with a GELU activation [22] is applied. Both sub-layers are wrapped with a skip-connection followed by a LayerNorm [10].

Training objective: Training BERT consists of two stages: pre-training and fine-tuning. Pre-training consists of two semi-supervised tasks. The first is masked language modeling (MLM) - prediction of randomly masked tokens in a sequence, where

15% of its tokens are masked and the model is asked to predict the indices of the original tokens; out of the 15% tokens, 80% of the time a token is replaced with a special [MASK] token, a random token 10% of the time, and left unchanged token 10% of the time [3]. Masking the target words with a special [MASK] token enables processing a bidirectional context without leaking information between the layers. Additionally, masking introduces noise to sequences and makes the prediction of the target words more difficult.

The second pre-training objective is the binary Next Sentence Prediction (NSP). An input sequence of BERT consists of two segments A and B, 50% of time B is the actual segment that follows A in the corpus and 50% of time it is a random segment. BERT is asked to predict whether A and B are consecutive segments. Note that this part of the objective uses only [CLS], thus affects the embedding vectors and the Transformer stack solely via the gradients through [CLS], which contributes to the unique features of [CLS] (such as its large magnitude).

During fine-tuning, a classification layer is added on top of the encoder stack and a task-specific objective function is used to train the classifier and the Transformer stack. The input pipeline for BERT consists of tokenizing words into wordpieces [23], and adding token, position, and segment embeddings to construct a fixed-length representation. Special [CLS] token is added at the beginning of the input and used for classification predictions in downstream tasks; special [SEP] token is used to separate input segments. Yang et al. [12] pointed out the issue of input inconsistency between pre-training and fine-tuning, as the [MASK] token is usually not used in downstream tasks. We observed a different discrepancy between the pre-training and fine-tuning versions of the model. Specifically, during pre-training a language modeling head, a classifier layer, is used on top of the stack of transformer blocks; later, the language modeling head is discarded and the rest of the model is used for fine-tuning on downstream tasks. Therefore, assuming that BERT indeed captures knowledge, it needs to be reflected in the encoder stack to be directly transferable to the downstream tasks.

IV. Language Modeling Head and Knowledge Retrieval in BERT

A hallmark feature of BERT is that using a masked language model makes computing bidirectional contexts possible. During pretraining, BERT is trained to learn how to replace [MASK] (80% of the time) and other random tokens (10% of the time) with the correct tokens. Since BERT pretraining includes a language model head (LM head) that is discarded when fine tuning downstream applications, it is important to understand if the encoder stack itself learns to perform token replacement. Our experiments and analyses show that it is not the case. To further understand BERT's ability for knowledge retrieval, we dissect the LM head to understand the underlying mechanisms. Our results show the LayerNorm plays an essential role in knowledge retrieval.

A. Knowledge Retrieval Using BERT

According to the masked language modeling framework, the common natural language answer retrieval approach is set up as follows. First, a sentence is extracted from the natural text resources on the Web, such as encyclopedias, manuals, newspapers, books, and so on. Second, the sentence is split into tokens from a model's dictionary. Finally, one of the tokens is replaced with the special [MASK] token to designate the subject of the query. The language modeling head is essentially a classifier that has been trained with the objective of minimizing the masked language modeling loss [3]. Intuitively, the loss rewards correct predictions of tokens that have been replaced with a [MASK] token in the training dataset during pre-training. Therefore, by looking at a few largest logits in the output position corresponding to the [MASK] token, we are able to retrieve the top candidates for the token that the pre-trained model considers to most likely appear, instead of the [MASK] token, in the original input sentence. Also, for each candidate, there is a prediction confidence score, produced by feeding the logits into softmax. While the classifier head accounts for a relatively tiny number of parameters, compared to the base BERT model size (only about 0.57% of the parameters of the entire model[2]), using the output from the encoder stack to predict the tokens directly (by bypassing the processing steps in the LM head) returns dramatically different results.

For example, we used the sentence *'[CLS] london is the capital of great britain. [SEP]'* as input and ran a few iterations of reconstruction. After the first iteration, the token sequence is transformed into *('[CLS]', '[CLS]', '[CLS]', '[CLS]', 'capital', '[CLS]', 'ned', '[CLS]', '##ann', '##vis')*, and on subsequent iterations, the token sequence converges to a fix-point *('[CLS]', '[CLS]', '[CLS]', '[CLS]', '[CLS]', '[CLS]', '[CLS]', '[CLS]', '[CLS]', '[CLS]')*. We observed such convergence on a few different input sequences and found the pattern worth studying. We believe the true reason for its emergence is the magnitude of [CLS] token being the largest across all tokens. Indeed, when we swap the magnitude of [CLS] and 'the' without changing their directions, after the first iteration, the same input token sequence is transformed into *('##city', 'london', 'the', 'the', 'capital', 'the', 'ned', 'britain', '##ann', '##vis')*, and on subsequent iterations, the token sequence converges to a fix-point *('the', 'the', 'the', 'the', 'the', 'the', 'the', 'the', 'the', 'the')*. We note that after swapping magnitudes, the 'the' token has taken the place of '[CLS]' token as being the point of convergence, as we would expect if the magnitude of the embedding played more important role than its direction. The role of embedding vector magnitude can also be expected due to inner products used in the reconstruction process.

From this study, we found that the bare encoder stack is providing unstable predictions, therefore, we argue that the LM head is very important for knowledge retrieval.

[2]BERT for MLM totals $109,514,298$ parameters, out of which $622,650$ are in the LM-Head.

B. LM Head Architecture

To understand the underlying mechanism for knowledge retrieval, we zoom into the language modeling head to track the roles of its components responsible for its ability to make sensible predictions. We have used the BERT provided by HuggingFace [24] as our implementation. The language modeling head consists of a sequence of modules: (1) fully connected layer taking H-dimensional input (*hidden dimension* parameter in BERT's configuration, equal to 784 for base uncased version) and producing H-dimensional output; (2) Gaussian error linear unit (GELU) nonlinearity [22]; (3) LayerNorm [10] with elementwise affine application; and (4) fully connected layer taking H-dimensional input and producing V-dimensional output (*vocabulary size* parameter in BERT's configuration, equal to 30522 for base uncased version) by computing inner products with the BERT word embeddings and adding bias that was learned during pre-training.

In the LayerNorm layer, for input vector x,

$$\mathtt{LN}(x) = \frac{x - mean(x)}{std(x)}. \tag{1}$$

In a variant with elementwise affine application, shared weight vector w and bias vector b (having the same dimension as the input vector) among all input samples are applied as

$$\mathtt{LNAff}(x) = \mathtt{LN}(x) \otimes w \oplus b, \tag{2}$$

where $\otimes$ is elementwise multiplication and $\oplus$ is elementwise addition. To study the inner structure of the language modeling head, we separate the fully connected layers and the LayerNorm module into a sequence of single operations. More specifically, we separate the fully connected layers into weight multiplication and bias addition and the LayerNorm module into LayerNorm normalization only (without elementwise affine application), elementwise weight multiplication ($\otimes$), and elementwise bias addition ($\oplus$). While such splitting reduces the inference speed, it allows us to analyze the intermediate results with a finer precision.

C. LM Head Activation Analysis

We have shown that LM Head is necessary for meaningful knowledge retrieval. Here, we analyze each of LM Head components in order to identify the root of this phenomenon, which should enable conscious usage of BERT in tasks requiring knowledge. In order to do that, we convert a sample sentence to a sequence of tokens and pass it as input to BERT with LM head. We then analyze the activations in the [MASK] token position via each LM head component. To estimate the original token reconstruction ability of these activations, we make a conceptual shortcut to the final head layer, computing the inner products of that H-dimensional activation with each of the V token embeddings, and measure the similarity of the V-dimensional result with the V-dimensional vector produced by BERT with entire LM head in the [MASK] token position. Since the indexes of the largest components of the V-dimensional vector are used to retrieve the most

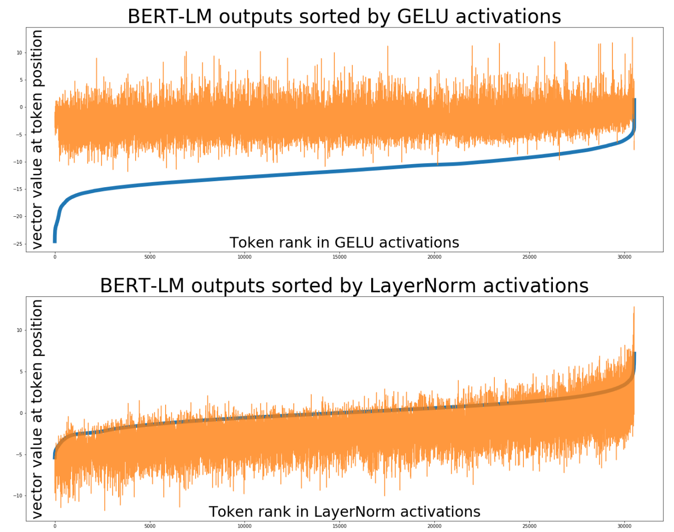

Fig. 2: BERT with LM head outputs (orange line) sorted by inner products of token embeddings with LM head components' activations (thick blue line; top: GELU, bottom: LayerNorm). Intuitively, the predictions are similar if the trends produced by orange and thick blue lines are similar. The trends visually align after the LayerNorm sublayer, as we can see in the bottom panel, and it coincides with reconstruction quality increase that we observe.

likely candidates for the reconstruction from the vocabulary, the absolute values of the components do not matter as much as their sorted order. Hence, we use Spearman's rank-order correlation [25] as the similarity metric. High similarity would indicate the LM head component responsible for high prediction quality. As a concrete example, consider the following probe suggested in [7]: *'Francesco Bartolomeo Conti was born in [MASK].'*. The *[MASK]* token here designates an unknown entity we are trying to retrieve from the pre-trained BERT. As BERT is pretrained to replace [MASK] by the correct token, [MASK] is the natural choice.

We find that while the output of language modeling head is only weakly uncorrelated with the inner products between the BERT last transformer layer and word embeddings, the strong correlation appears after the LayerNorm step. In Figure 2, we can see that the ranks of intermediate representations start agreeing with the ranks of final representations precisely after LayerNorm activation sublayer. For the example, the top predictions obtained using the LayerNorm activations are 'turin', 'rome', 'bologna', 'pisa', and 'milan', and the ones obtained from GELU layer (immediately before the LayerNorm step) are are '[CLS]', '[MASK]', '[SEP]', 'there', and 'here'. It is evident that *the LM head is driven into the correct answer's context after the LayerNorm layer*. In particular, we analyzed the LayerNorm weights; while the details are given in Appendix (Figures 8 and 9), their effect on most of the input vectors is to multiply them by a factor of around 2.3, except for [CLS] and other tokens similar to [CLS] embedding vectors, whose magnitude is either reduced or multiplied by a factor of around

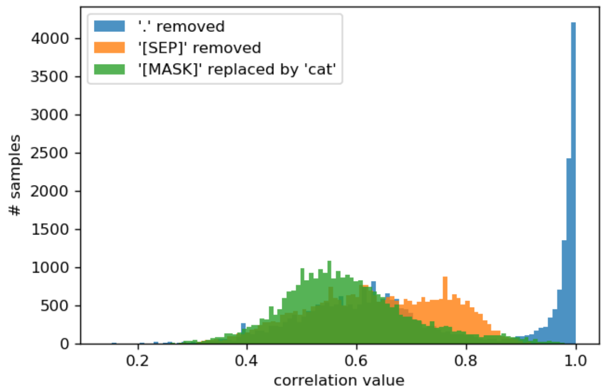

Fig. 3: Histogram of Spearman's rank-order correlations between logits of the original sample and logits of a perturbed sample, collected over entire ConceptNet dataset used in LAMA. The perturbations are (1) removing the '.' token ending the sample sentence or sentences; (2) removing the [SEP] token ending the token sequence serving as input to BERT; (3) replacing the [MASK] token with the 'cat' one. The decrease in reconstruction quality between perturbation strategies is reflected in the correlations decrease as the bell curve shifts to the left.

1.5. Therefore, the ranking orders are changed significantly.

In order to study LM head components' impact on unmasking words systematically, we use the ConceptNet dataset provided with the LAMA probe [7]. It consists of 29774 samples; we used the following fields: (1) a sentence with '[MASK]' replacing the word that appeared in the original text; (2) the ground truth token that we use to estimate the reconstruction accuracy. We study embeddings produced by each LM head component by looking into logits in the output sequence position of [MASK] token; to assess the similarity between each of these embeddings and LM head output, we compute Spearman's rank-order correlation. The average components' histograms of these correlations over the entire ConceptNet dataset are in Figure 1. It shows that the similarity significantly increases on average after the LayerNorm component.

V. Replacement Token Analysis

Here we further investigate the generalization and robustness of BERT models for knowledge retrieval. BERT is pretrained to learn how to replace [MASK] 80% of the cases and 10% of them to replace inconsistent tokens. As pointed out by Yang et al. [12], downstream applications do not use [MASK] typically and it is important to understand how the models generalize to other input sequences. We consider two cases that the models should handle well in order to be useful for downstream applications. The first one is the effect of single word replacement on the retrieval results and the second one is the effect of inconsequential syntactical changes on the results. Note that many more similar and other changes can be designed and studied. For example, consider the probe *'Barack Obama was born in X.'*, where we can vary the token *X* to investigate how the predictions depend on the replacement token itself. When we set *X* to *'ukraine'*, the top prediction is *'ukraine'* with the highest confidence, followed by *'ukrainian'*, *'where'*, *'crimea'*, and *'delta'*, *'kiev'*. We observe this pattern with a number of probes and replacement tokens. To track the replacement token effect on the prediction quantitatively, we fixed a probe sentence and ran a prediction for each token in the vocabulary used as the replacement token. In approximately 75% of cases, the replacement tokens themselves got predicted as the top prediction. Table II shows the first few most frequent tokens reconstructed in a probe where the '[MASK]' token is being substituted by each token in the vocabulary. As we discover, the factually correct answer (**florence**) is *not* the most popular one, and we can see many tokens in this list have a common *theme* (Italy). Also, in the substantial majority of cases, the replaced token reconstructs itself, thus making queries fragile. Furthermore, we also ran experiments by replacing [MASK] using particular tokens on the entire ConceptNet dataset. The results show that the replacement tokens themselves are retrieved around 73% of time (from 65% to 84%). This experiment demonstrates that while the influence of the context on the output representations improves downstream classification tasks and other applications, it is not sufficient for the model to retrieve knowledge reliably and generalize well to new sentences in meaningful ways.

Furthermore, we quantify the model fragility by crafting perturbations that can be applied to a large number of samples subject to token unmasking. We considered three shallow changes: (1) removing the period ending the sentence (if there is one), (2) removing [SEP] token normally inserted at the end of input token sequence, and (3) replacing [MASK] token with 'cat' while still using ranks of output components in the token position to retrieve the reconstructed token from vocabulary. While such changes are minimal in terms of edit distance [26] and do not change the semantics of input sentences, they affect the predictions significantly both in individual samples and cumulatively. For example, consider the sentence *'One of the things you do when you are alive is [MASK].'* taken from the ConceptNet dataset [7]. The top 5 original predictions of uncased base BERT configuration are *'sleep', 'think', 'cry', 'eat', 'pray'*. With the final period token removed from the original token sequence, the top predictions become *'.', ';', '?', '!', '|'*; with [SEP] token removed from the original token sequence – *'.', '"', 'the', ',', ')'*; with [MASK] token replaced by the 'cat' one – *'cat', 'cats', 'dog', '##cat', 'kitten'*. We systematically evaluated the three changes. In the original cases and the first two cases, we breakdown cases based on whether the [MASK] token is at the end (meaning it is the last token other than '.', and [SEP]) or not. Table I displays evaluation of retrieval quality of the original and perturbed samples on the entire ConceptNet dataset. It shows that the perturbations decrease the fraction of samples where the correct token is in top N predicted tokens according to logit ranking for each N in (1, 2, 5, 10, 20, 50, 100), and the decrease is at least two-fold. The results show clearly syntactic patterns play an

PROBE TYPE	C@1	C@2	C@5	C@10	C@20	C@50	C@100	Total
original	3710	5285	7743	9460	11219	13825	15843	29774
- [MASK] at the end	1216	1822	2934	3839	4753	6387	7697	18796
- other	2494	3463	4809	5621	6466	7438	8146	10978
ending `.' removed	2350	3327	4588	6120	8798	12500	14820	29714
- [MASK] at the end	6	22	111	766	2606	5214	6806	18780
- other	2344	3305	4477	5354	6192	7286	8014	10934
[SEP] removed	1009	1649	2612	3564	4632	6399	8621	29774
- [MASK] at the end	52	175	387	608	922	1609	2773	18796
- other	957	1474	2225	2972	3710	4790	5848	10978
[MASK] / cat	702	1373	2583	3566	4722	6620	8470	29774

TABLE I: Retrieval counts for BERT base uncased on ConceptNet dataset. *Count at K* is the modification of *hits at K* metric, showing the count of samples, where the correct answer appears in the list of K top candidates. The *Total* column shows the total number of samples in each probe category.

Token	rome	bologna	genoa	ca	verona	como	**florence**	c	venice	town	padua	villa
Count	2519	685	586	391	363	320	**273**	186	152	95	81	75

TABLE II: Comprehensive token substitution exploration. We take the sentence *'Francesco Bartolomeo Conti was born in X.'*, substitute X with each token in the vocabulary, and ask BERT with a LM head to guess which original token was substituted. In the majority of cases, X and predicted token turn out to be the same. In the remaining 7608 out of 30522 cases, they are not the same, and the most frequently predicted tokens in this group are in the table above. We can see that the Italian cities prevail, and the ground truth answer from the world knowledge point of view (Florence) has a smaller number of predictions than a few other tokens.

important role in the knowledge retrieved. The experiments show that having [MASK] in the third to last position in the input token sequence accounts for the overwhelming majority of mispredictions both in final period removal and [SEP] removal probes. It is likely due to that the words tend to co-occur much more strongly with their neighboring words [27], a change closer to [MASK] would therefore have a stronger effect. Finally, this evaluation shows that there is a clear difference in the effect among the perturbations. Removing the final period is the least confusing while token replacement by 'cat' causes the most confusion. We also experimented with tokens other than 'cat'; the performance is generally similar. Using 'thereof' reduces the performance even more with 316 for C@1 and 1889 for C@10 respectively.

We also compute a histogram of Spearman's rank-order correlations [25] between the original and each of the perturbed cases' output vectors to probe whether the vectors' geometry affects the prediction results. The results are displayed in Figure 3. We observe that the pattern with prediction quality deterioration is reflected also in the correlation plots. While removing '.' still correlates very strongly with the original predictions, the correlation is reduced substantially when the [MASK] is replaced by 'cat'. Additional examples are also available in Appendix (Figure 7).

VI. Dangerous Factual Errors

In the previous sections, we have explored structural limitations of BERT to alter its knowledge retrieval performance. Here, we aim to highlight a limitation of another kind: due to its MLM objective, BERT is trained to reconstruct a token that is statistically likely to appear, whether a sentence is factually correct or not. For instance, for the input sentence *"Vaccines cause [MASK]."*, the top 5 predictions are *'cancer', 'disease', 'death', 'diseases', 'mortality'*; more examples can be found in Table IV. Such factual misconceptions could cause a disruption in any application that uses a pre-trained model as the source of knowledge.

In order to estimate the possible degree of confusion, we have collected a dataset with common misconceptions. During construction of the dataset, we have used the misconceptions infographic[3] with 100 widely believed incorrect facts. However, we had to reword the sentences, so that there is a blank that could be filled in for each sentence. We have run an open-ended experiment, where any token from the vocabulary can replace the blank, and its output logit is used for ranking. We have also run an experiment with a *limited* number of replacement choices to mimic tests used for human evaluation, where only a few options are available for a given probe. For example, there is a common misconception that rust causes tetanus; the relation is at most correlative, as it is not feasible to distinguish between tetanus-prone and non-tetanus-prone wounds [28]. In the limited-choice blank replacement task, the prompt is *'Rust [MASK] directly cause tetanus.'*, and the options are *'can'*, *'also'*, *'will'* and *'cannot'*; only the last option is correct. There are 109 samples in our misconception dataset. The performance on the dataset was compared among BERT and its three variations: RoBERTa [4], ALBERT [29], and ELECTRA [30]. The results are shown in Table III. None of the models achieve an accuracy much higher than chance, with BERT being the best one with an accuracy of 34.86%. Furthermore, error analysis shows that most of the mispredictions, about 70% for all the models, correspond to the answers most directly reflecting the common misconceptions. Somewhat surprisingly, the accuracy of RoBERTa is about 3.5% lower than BERT, while the number of answers reflecting the common mispredictions is 1.5% higher.

[3]https://www.geekwrapped.com/myth-infographic

	Acc	MC	C@1	C@2	C@5	C@10	C@20	C@50	C@100
RoBERTa	31.19	73.33	15	23	37	53	64	76	85
ALBERT	33.94	70.83	8	16	28	46	59	72	82
ELECTRA	30.28	67.10	11	21	31	51	60	81	89
BERT	34.86	71.83	14	22	33	57	66	85	91

TABLE III: Results of the misconception experiment. MC stands for the percent of errors that most directly correspond to the common misconceptions. There are 109 samples in the dataset.

Query	Answers
The light bulb was [MASK] invented by Thomas Edison.	**A. first** B. original C. science <u>D. not</u>
Tomatoes are a [MASK].	**A. vegetable** B. meat C. juice <u>D. fruit</u>
Schizophrenia is [MASK] to dissociative identity disorder (having multiple personalities).	**A. equivalent** B. ill C. forget <u>D. different</u>
Ostriches stick their heads in the sand to [MASK].	**A. hide** B. cover C. threat <u>D. dig</u>

TABLE IV: Examples of queries from the misconception dataset. Answers predicted by BERT are in bold, correct answers are underlined.

The main differences between BERT and RoBERTa are that the latter was trained on a larger corpus and for a larger number of steps, resulting in a larger exposure to "common knowledge" from the Web based resources, which are not always reliable. The results suggest the obvious that accumulating more data will not lead to more reliable knowledge and using larger datasets will not fix the inherent issues.

VII. Discussion

By the masked language model, BERT produces representations that depend on the entire input sequence. Empirical results on benchmarks demonstrate the effectiveness of bidirectional contexts. One consequence is that the resulting model is more difficult to analyze, as its outputs heavily depend on the input sequences themselves. While the model should have learned that [MASK] tokens need to be replaced in the input sequences, it is not clear what the model would do when there are no [MASK] tokens. The uncertainty introduced by using random words for the 10% of the replacements renders the situation more complex, as it is constantly unclear which words need to be replaced, on top of the [MASK] tokens. Ideally, the model should first learn if the input sequence is consistent; if it is not consistent, it should then identify the words that need to be replaced. Unfortunately, during pre-training consistent inputs are non-existent. Furthermore, due to its continuous nature, the model can not avoid the influence of the random or [MASK] tokens on the output representations, therefore, cannot make consistent decisions. As a result, given 'Barack Obama was born in X.', the model will likely produce a set of different answers depending on X. Ironically, the model will change the correct answer to other ones influenced by the replacement token. We argue that one should not interpret occasional correct statements of BERT as knowledge sources, which must be robust against noise and can not change due to external inputs. At a minimum, knowledge sources can not give answers that are constantly changing, which is what BERT models do.

VIII. Conclusion and Future Work

As the predictions in BERT models depend on the input sequence itself, the knowledge stored in BERT is input-dependent and therefore not reliable. In this paper we have performed a number of experiments confirming that. We have explored two approaches to show fragility. The first is making minor syntactic changes that *systematically* perturb the inference results, and the latter is to exploit *the high frequency of incorrect facts*. The results show that one should proceed with caution when using pre-trained language models as source of knowledge in downstream applications, which may include search engines and expert systems.

Besides the two perturbation types we have used, many interesting ones can be designed and studied to reveal the knowledge retrieval properties of BERT. For example, one could reorder sentences based on parse trees without changing the semantics. As BERT contextual representations tend to depend heavily on words in a local neighborhood [27], reordering should influence the knowledge retrieval results and the actual impact can be modeled empirically. Another interesting type is to modify sentences based on their inherent ambiguities and redundancies. Further promising future work directions include developing automated methods for debiasing language models from most commonly held misconceptions, including ones leading to gender and racial biases, detecting such misconceptions, and making the language models more robust to query syntax changes.

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